

Euclidean Geometry In Mathematical Olympiads

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Problem 1.33



trivial

Chapter 1

fgm

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#1

Problem 1.33 Let ABC be a triangle and let ray AO meet BC at D' . Point K is selected so that KA is tangent to (ABC) and $\angle KC = 90$. Prove that $KD' \parallel AB$.

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#2

Solution



The problem is near trivial. First, notice $\angle KAC = \angle ABD'$. Also, notice $AD'CK$ is cyclic, so $\angle KAC = \angle KD'C$. The result follows.

Drunken_Master

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#3

As $KAD'C$ is cyclic $\angle AKD' = \angle C$ and $\angle BAK = 180^\circ$, so $AB \parallel KD'$.

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#4

Angle KC refers to angle KCB as opposed to angle AKC, I assume?

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#5

wait, how is BAK 180
it certinly doesnt look like that on my diagram

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#6

gimme help i am not good

Kuroshio

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#7

$\angle D'AK = 90^\circ = \angle D'CK$ (tangency)
Points A, K, C, D are *conyclic*.
 $\angle KD'C = \angle KAC = \angle ABC$ (alternate segment theorem)
It follows that $AB \parallel KD'$

This post has been edited 1 time. Last edited by Kuroshio, Jul 16, 2020, 4:44 PM

Kgxtixigct

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#8



Kuroshio wrote:

$\angle D'AK = 90^\circ = \angle D'CK$ (tangency)
Points A, K, C, D are con cyclic.
 $\angle KD'C = \angle KAC = \angle ABC$ (alternate segment theorem)
So $AB \parallel KD'$